INCH-POUND
MIL-DTL-24646/15C
w/AMENDMENT 1
2 October 2024
SUPERSEDING
MIL-DTL-24646/15C
29 January 2015

DETAIL SPECIFICATION SHEET

FUSE BOXES, FEEDER JUNCTION, SUBMERSIBLE, (15 FEET), 250 VOLTS, ALTERNATING CURRENT, FEED THROUGH TWIN CONDUCTORS, SINGLE AND THREE PHASE BUSES, SINGLE AND THREE PHASE BRANCHES, 0-30 AND 31-60 AMPERE FUSES, SYMBOLS 685.1, 686.1, 691.1, AND 692.1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24646.

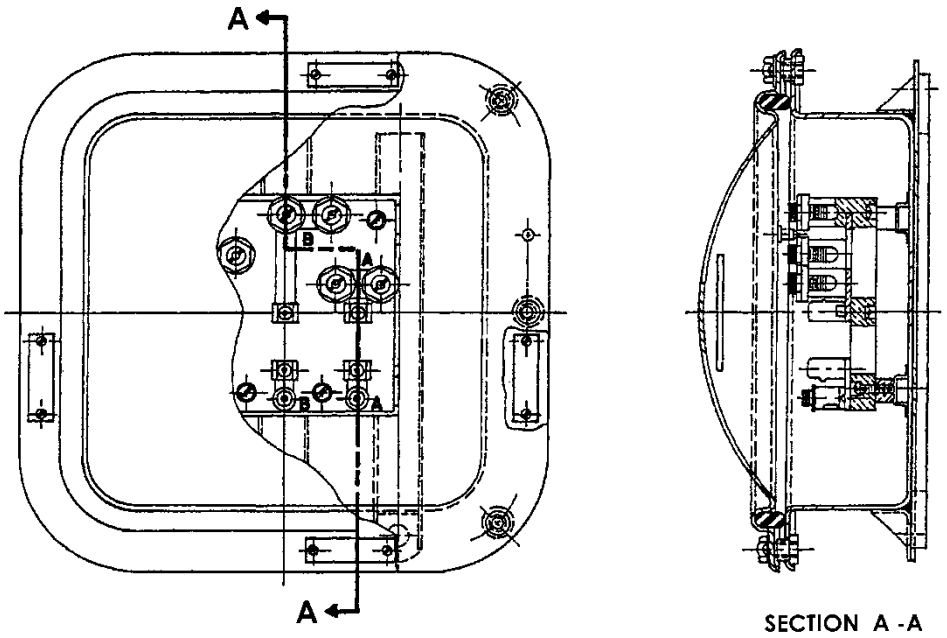


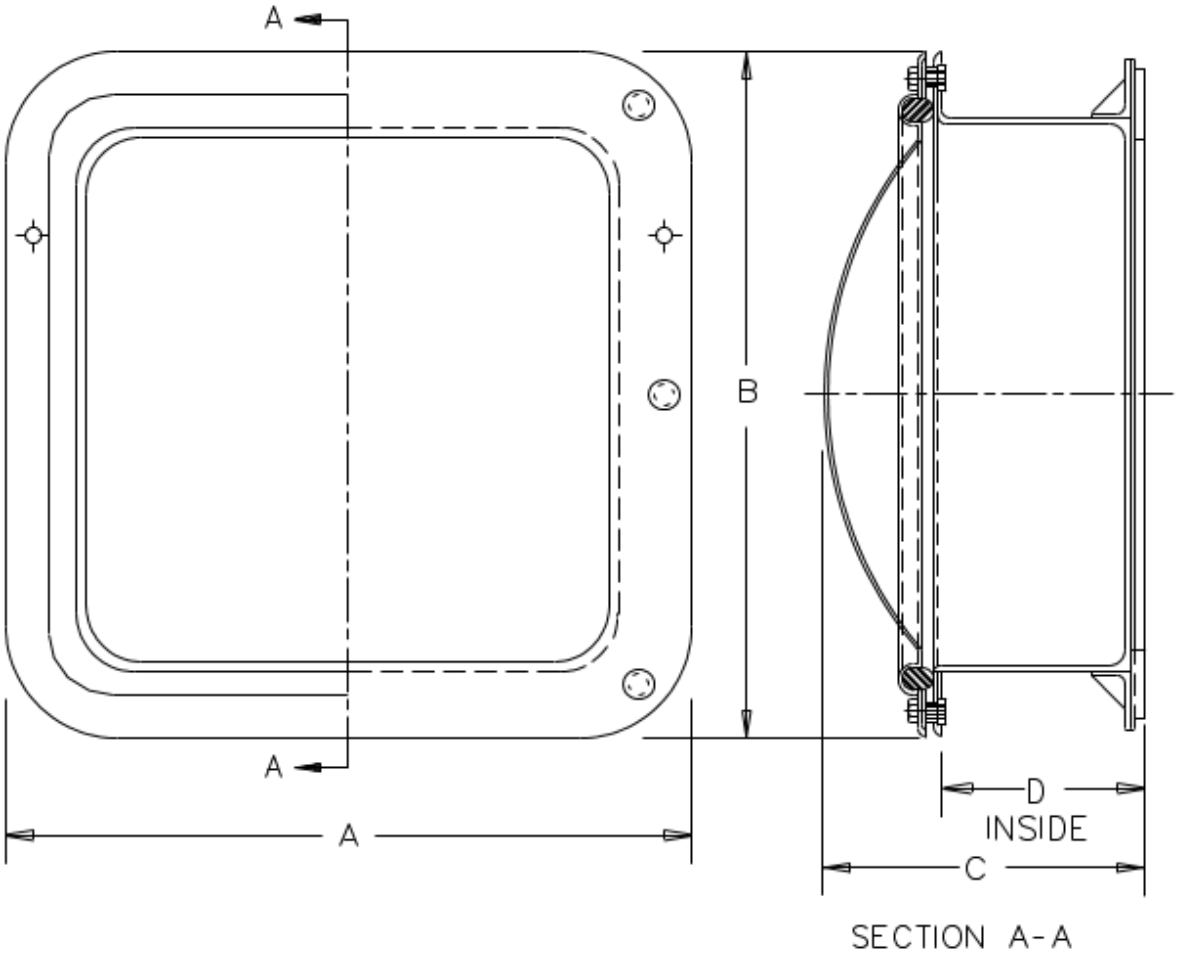
FIGURE 1. Fuse box assembly, typical. Item No. 01, 02, 03, 04
(Symbol 691.1 shown)

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified herein, tolerances shall be as follows:
   Two place decimals - plus or minus 0.02.
   Three place decimals - plus or minus 0.005.

AMSC N/A
FSC 5920

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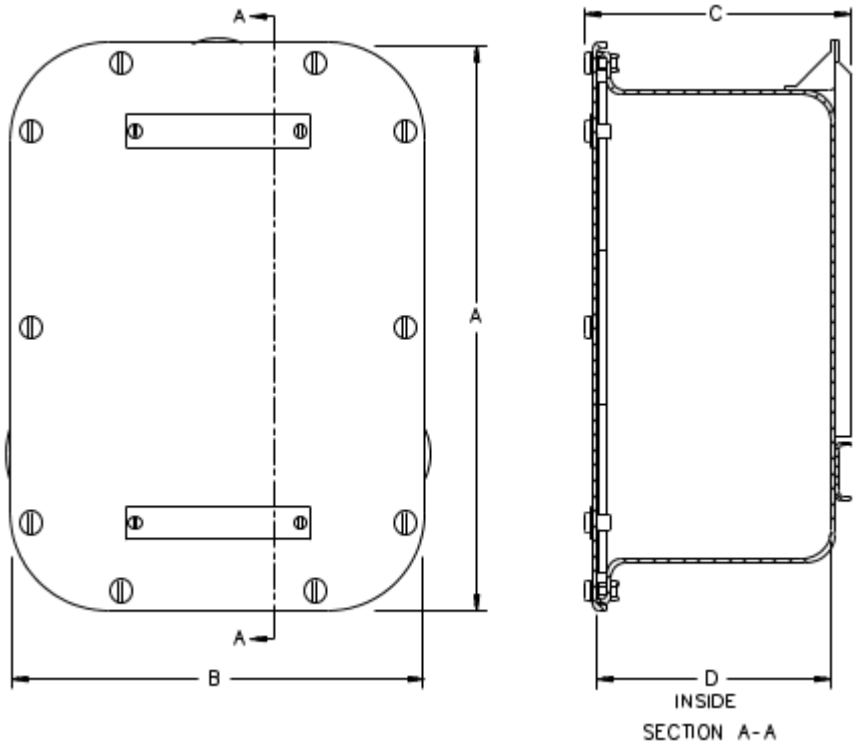
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(Symbols 691.1, 692.1)

FIGURE 2. Enclosure, overall dimensions. Item No. 05, 06, 07
(Symbols 685.1, 686.1, 691.1, and 692.1)

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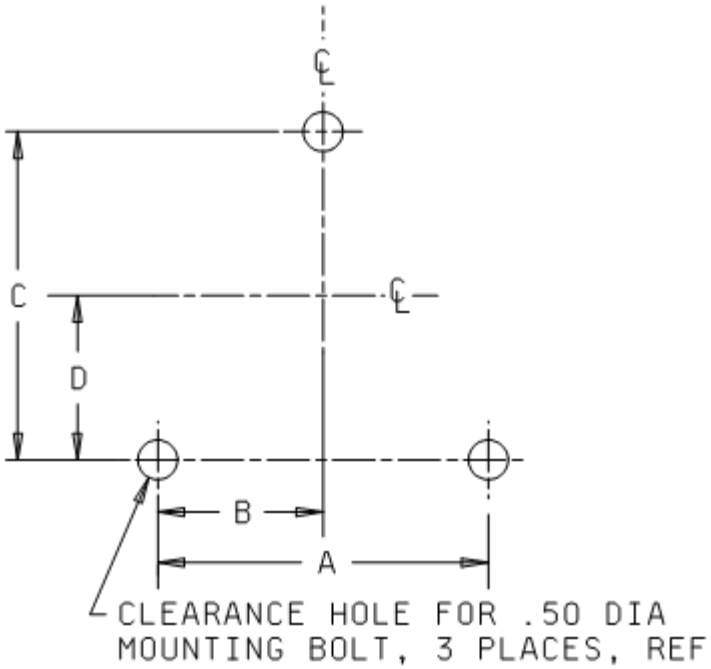


(Symbols 685.1, 686.1)

SYMBOL NO	DIMENSIONS (REF)			
	A	B	C	D
685.1	10.94	7.94	5.12	4.50
686.1	10.94	7.94	5.12	4.50
691.1	13.38	13.38	5.62	3.00
692.1	15.38	16.38	7.62	5.00

FIGURE 2. Enclosure, overall dimensions - continued. Item No. 05, 06, 07
(Symbols 685.1, 686.1, 691.1, and 692.1)

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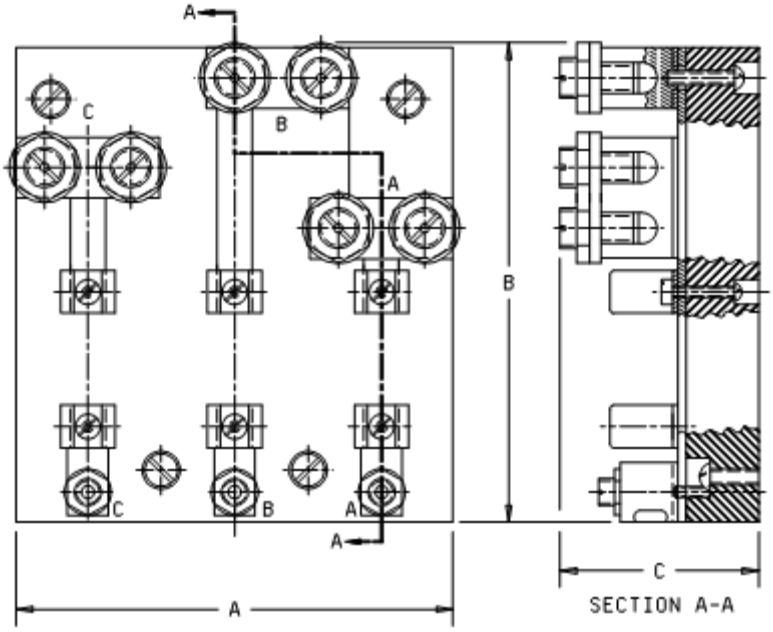


For symbols 691.1 and 692.1. For symbols 685.1 and 686.1 a clearance hole for .375 dia. mounting bolt is needed.

SYMBOL NO	DIMENSIONS			
	A	B	C	D
685.1	7.25	3.62	7.88	2.75
686.1	7.25	3.62	7.88	2.75
691.1	5.75	2.88	11.44	5.72
692.1	9.25	4.62	13.44	6.72

FIGURE 3. Mounting dimensions, ref.
(Symbols 685.1, 686.1, 691.1 and 692.1)

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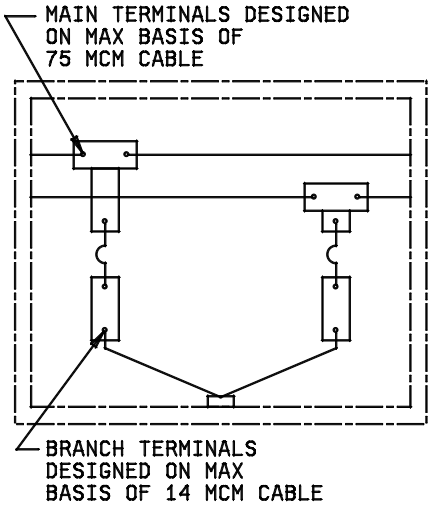


SYMBOL NO	DIMENSIONS (MAX)		
	A	B	C
685.1	6.25	4.63	2.06
686.1	6.25	4.63	2.06
691.1	5.50	5.63	2.50
692.1	8.38	6.88	2.50

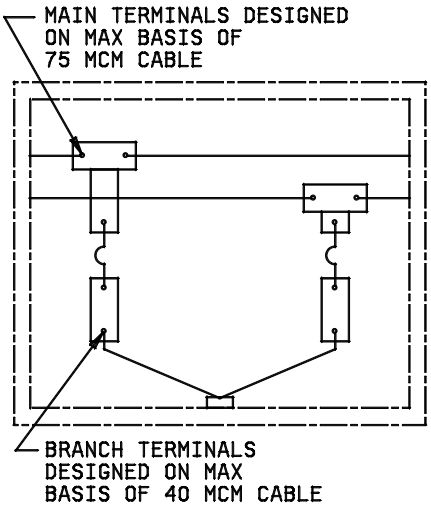
FIGURE 4. Fuse block assembly, typical. Item No. 08, 09, 10, 11
(Symbol 691.1 shown)

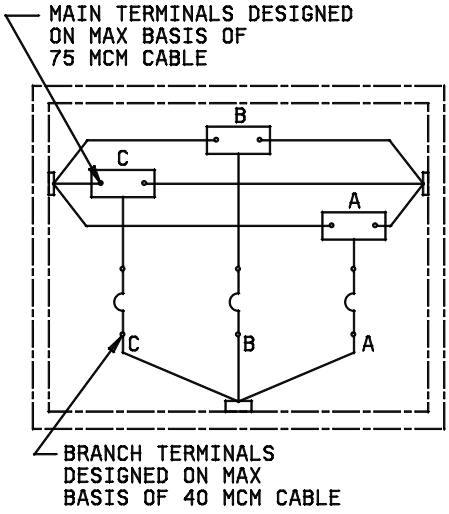
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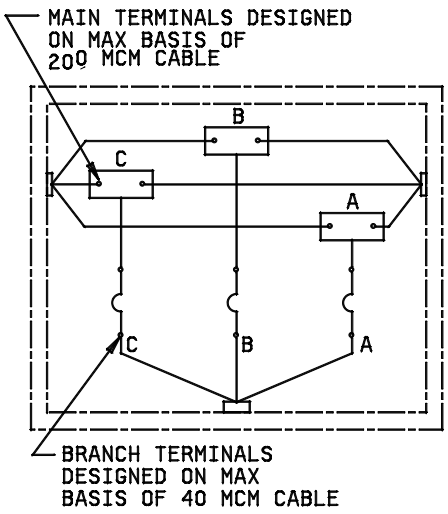
A. One single-phase branch
(Symbol 685.1)



B. One single-phase branch
(Symbol 686.1)







C. One 3-phase branch
(Symbol 691.1)

D. One 3-phase branch
(Symbol 692.1)

FIGURE 5. Wiring diagrams. 12, 13, 14, 15

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REQUIREMENTS

Symbol numbers: See table I.

Design:

Electrical rating: 250 volts alternating current, 0-30, and 31-60 ampere, 60 hertz.
Dimensions and configuration: See figures 1, 2, 3 and 4.
Wiring diagrams: See figure 5.
Mounting dimensions: See figure 3.
Degree of enclosure: Submersible (15 feet).
Designation and marking: See MIL-DTL-24646.
Parts: The parts listed in table II and specified in MIL-DTL-24646 shall be used in the construction of the fuse boxes specified herein.

General information:
Military part numbers and general information is as specified in table I.

TABLE I. Military part numbers and general information.

Military Part no.	Symbol	Approx. Weight in pounds 1/	No. of circuits	Feeder MCM	Branch MCM	Fuse size	Bus bar size	Supersedes	
								Drawing No.	Type
M24646/15-001	685.1	10.0	1	75	14	0-30A	0.75 x 0.12	9000-S-6202-74183	F-47A
M24646/15 002	686.1	10.0	1	75	40	31-60A	0.75 x 0.12	9000-S-6202-74185	F-48A
M24646/15-003	691.1	22.0	1	75	40	0-30A	0.75 x 0.12	9000-S-6202-74189	F-25A
M24646/15-004	692.1	30.0	1	200	40	31-60A	0.75 x 0.12	9000-S-6202-74190	F-27A

1/ The weight of these fuse boxes may vary + or - 15 percent.

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TABLE II. Parts list.

Item no. 1/	Description	QTY M /-				Material/ specification	Remarks
		001	002	003	004		
01	Fuse box assembly (sym 685.1)	-				MIL-DTL-24646/15	See figure 1
02	Fuse box assembly (sym 686.1)		-			MIL-DTL-24646/15	See figure 1
03	Fuse box assembly (sym 691.1			-		MIL-DTL-24646/15	See figure 1
04	Fuse box assembly (sym 692.1)				-	MIL-DTL-24646/15	See figure 1
05 2/	Enclosure (6 x 9 x 4.5)	1	1			M24142/3-001	See figure 2
06 3/	Enclosure (10 x 10 x 3)			1		M24142/6-003	See figure 2
07	Enclosure (12 x 13 x 5)				1	M24142/6-008	See figure 2
08	Fuse block assembly	1				MIL-DTL-24646/15	See figure 4
09	Fuse block assembly		1			MIL-DTL-24646/15	See figure 4
10	Fuse block assembly			1		MIL-DTL-24646/15	See figure 4
11	Fuse block assembly				1	MIL-DTL-24646/15	See figure 4
12	Wiring diagram	1				MIL-DTL-2036	See figure 5
13	Wiring diagram		1			MIL-DTL-2036	See figure 5
14	Wiring diagram			1		MIL-DTL-2036	See figure 5
15	Wiring diagram				1	MIL-DTL-2036	See figure 5
16	Fuse clip (0-30A)	4	-	6	-	A-A-55507/1-06	N/A
17	Fuse clip (31-60A)	-	4	-	6	A-A-55507/1-08	N/A
18	Main terminal assembly	2	2	3	3	NAVSEA/SHIPS 9000-S-6202-73246	N/A
19	Identification plate	1	1	1	1	Brass, ASTM B36/B36M B121/B121M	N/A
20	Information plate	1	1	3	3	Brass, ASTM B36/B36M B121/B121M	N/A

1/ Item numbers after the figure titles are the same as the item numbers shown in this table.
2/ Enclosure is same as MIL-E-24142/3, except for inside depth.
3/ Enclosure is same as MIL-E-24142/6, except for inside depth.
4/ A-A-55507/1-06 and A-A-55507/1-08 come from document A-A-55507/1

Referenced documents. In addition to MIL-DTL-24646, this document references the following:
MIL-DTL-2036 MIL-E-24142/3 MIL-E-24142/6 ASTM B36/B36M ASTM B121/B121M
NAVSEA/SHIPS 9000-S-6202-73246

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(Project 5920-2024-033)

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